WORLD MISSION
HIGH SCHOOL

60
DECISION

EXAM DETAILS		TRAINEE'S DETAILS	
Sector	ICT & MULTIMEDIA	Trainee's name	
Trade	SOD	Trainer's name	H. Love
Module code	SWDVC301	Module title	Version Control System
Level	L3 SOD	Date	December ….., 2025

Instructions

*1. This question paper contains only **Five** sections. Answer all questions*
*2. You have **1 hour 30'** to complete the examination*
3. For multiple choices Questions, True/False and Matching Questions use this paper for answers. If Not, Answer will be ignored at all.

Section A: (15 marks) Multiple Choice Questions (Select by **circling** the best answer option)

1. What is the main role of **Version Control**?
   A. To manage the operating system's command line interface.
   B. To provide a web-based platform for social coding.
   C. To track and manage changes made to a project's source code over time.
   D. To automatically build and deploy code changes to production systems.
2. Which company(who) created Git in 2005?
   A. Microsoft
   B. GitHub
   C. Matt Mackall
   D. Linus Torvalds
3. Which of the following is **NOT** listed as a benefit of using version control systems?
   A. Backup and Disaster Recovery
   B. Continuous Integration and Deployment
   C. Increased Hardware Processing Speed
   D. Accountability
4. In a **Local Version Control** system, where are all the patches and file changes stored?
   A. On a central, shared repository server.

B. On a developer's local machine as a local database.

C. On a remote server like GitHub.

D. Only on the server responsible for Continuous Integration.

5. What is the major vulnerability of a **Centralized Version Control System (CVCS)**?

   A. It requires more disk space than other version control systems.

   B. Collaboration with other developers is nearly impossible.

   C. It is vulnerable to a single point of failure (if the central server goes down).

   D. Developers must work offline and cannot access the complete codebase.

6. Which of the following is the most well-known example of a **Distributed Version Control System (DVCS)**?

   A. SVN (Subversion)

   B. CVS (Concurrent Version System)

   C. Microsoft Team Foundation Server (TFS)

   D. Git (Global Information Tracker)

7. Git is based on the concept of a **Repository**. What does a repository contain?

   A. Only the latest snapshot of the code, excluding history.

   B. The source code files, history, and metadata like who made changes.

   C. Only the executable build scripts and configuration files.

   D. Just the .gitignore file and user manuals.

8. In Git, what is a **Commit**?

   A. The process of combining changes from one branch into another.

   B. A separate copy of the codebase used for development purposes.

   C. A snapshot of the codebase at a particular point in time.

   D. The shared location where all team members can access the code.

9. Which Git concept represents a copy of the repository stored on a remote server like GitLab?

   A. Working Directory

   B. Branch

   C. Remote

   D. Staging Area

10. What is the process of downloading changes from a remote repository and combining them into the local codebase called?

    A. Pushing

    B. Committing

    C. Cloning

    D. Pulling

11. What is the purpose of the **Staging Area** in Git's architecture?

A. It is the directory where the developer edits and modifies files.

B. It stores all the changes made to the codebase over time.

C. It is where developers review and prepare specific changes before committing them.

D. It is the official location for the master branch.

12. The git init command is used for what purpose?

A. To add files to the remote repository.

B. To merge changes from one branch to another.

C. To install Git on your operating system.

D. To initialize a new Git repository in a project directory.

13. What is the primary purpose of a .gitignore file?

A. To specify which files should be tracked and included in version control.

B. To tell Git to ignore intentionally untracked files (e.g., generated files).

C. To configure a new Git repository's global user name.

D. To rename a remote repository URL.

14. GitHub is best described as:

A. A local version control system for individual projects.

B. An American company that only tracks bugs and issues.

C. An Internet hosting service for software development using Git.

D. A simple command-line interface for interacting with the operating system.

15. What is the purpose of the git clone <repository-url> command?

A. To add a new remote repository to your local Git configuration.

B. To upload your local commits to the remote server.

C. To create a local copy of a remote repository on your machine.

D. To show detailed information about the tracked remote repository.

SECTION B: MATCHING QUESTIONS (5Marks)

Match the command/concept in Column A with its most accurate description in Column B.

Answer	**Column A**	**Column B**
1=	**1.** git config	**A.** A file that uses patterns to exclude certain files from being tracked by Git.
2=.....	**2.** git remote	**B.** The process of creating a separate copy of the codebase for development/testing.

Answer	Column A	Column B
3= ……	**3.** .gitignore file	**C.** Used to set global or local user name and email settings for Git.
4= ……	**4.** Branching	**D.** Used to manage, view, add, rename, or remove remote repositories.
5= ……	**5.** git add	**E.** Used to track changes to source code and maintain a history of all modifications.
		F. Moves changes from the Working Directory into the Staging Area.
		G. A type of version control system where all file versions are on a single server.

SECTION C: TRUE/FALSE QUESTIONS (10Marks)

Instructions: State by selecting whether each statement is True or False.

1. GitHub provides a Web-based graphical interface. (True / False)
2. If the central server of a Centralized Version Control System (CVCS) goes down, developers can still access the code repository and collaborate freely. (True / False)
3. The main advantage of Distributed Version Control Systems (DVCS) is that they require less disk space than other version control systems. (True / False)
4. In the Git workflow, a Pull Request is typically created *after* changes have been merged into the main codebase. (True / False)
5. A rule starting with an exclamation mark (!) in a .gitignore file tells Git to include the matched files, negating a previous ignore pattern. (True / False)

SECTION D: SHORT ANSWER/DEFINITION(15Marks)

1. Define **Version Control** and explain one key reason it is a critical tool in software development. **(4marks)**
2. What is **GitHub**? Name one collaboration tool it provides. **(4marks)**

3. List the three main components of **Git's Architecture**. **(3marks)**
4. State the command used to check the version of Git installed on your machine and the command to view a list of all remote repositories associated with your local repository. **(4marks)**

SECTION E: PROBLEM-SOLVING/ANALYSIS (15Marks)

1. **Scenario: Remote Failure** (5Marks) A team of developers is using a version control system. Their main server suddenly experiences a hardware failure, and the central database containing the project history is temporarily inaccessible.
   - **Identify** which type of version control system the team is *most likely* using, based on this single point of failure. (1Mark)
   - **Explain** the four main consequence this failure has on collaboration and version saving for the team. (4Marks)
2. **Git Terminology: Commit vs. Branch** (5Marks)
   Distinguish clearly between a **Commit** and a **Branch** in the context of Git's basic concepts. How do they work together to track project progress?
3. **Scenario: Initializing a Project** (5Marks) A developer starts a new project in an empty folder called MyNewApp. They write some code and then decide they want to use Git to track their changes.

a. What is the **first Git command** the developer must run in the MyNewApp folder?(1 Mark)

b. **Explain** why this command is necessary and what it creates within the project directory. (4Marks)

END OF ASSESSMENT